循环语句

In [6]:
```
i = 1
while i<= 100:
    print(f"想你{i}")
    if i==20:
        break
    i +=1
```

```
想你1
想你2
想你3
想你4
想你5
想你6
想你7
想你8
想你9
想你10
想你11
想你12
想你13
想你14
想你15
想你16
想你17
想你18
想你19
想你20
```

In [17]:
```
i = 1
while i<=5:
    print('*-*',end='')
    i +=1
    print()
```

```
*-*
*-*
*-*
*-*
*-*
```

In [ ]:
```
i = 1
while i <= 5:
    s = 1
    while s <= 5:
        print('*',end='')
        s += 1
    print()
    i += 1
```

```
*****
*****
*****
*****
*****
```

In [24]:
```
i = 5
while i >= 0:
```

```
    s = 1
    while s <= i:
        print('*',end='')
        s += 1
    print()
    i -= 1
```

```
*****
****
***
**
*
```

In [ ]:

```
i = 1
while i <= 5:
    s = 1
    while s <= i:
        print('*', end='')
        s += 1
    print()
    i += 1
```

```
*
**
***
****
*****
```

In [ ]:

```
#打印乘法表
i = 1
while i <= 9:
    s = 1
    while s <= i:
        print(f"{i}*{s}  = {i*s}",end='\t')
        s += 1
    print()
    i += 1
```

```
1*1  = 1
2*1  = 2        2*2  = 4
3*1  = 3        3*2  = 6        3*3  = 9
4*1  = 4        4*2  = 8        4*3  = 12       4*4  = 16
5*1  = 5        5*2  = 10       5*3  = 15       5*4  = 20       5*5  = 25
6*1  = 6        6*2  = 12       6*3  = 18       6*4  = 24       6*5  = 30       6
*6  = 36
7*1  = 7        7*2  = 14       7*3  = 21       7*4  = 28       7*5  = 35       7
*6  = 42        7*7  = 49
8*1  = 8        8*2  = 16       8*3  = 24       8*4  = 32       8*5  = 40       8
*6  = 48        8*7  = 56       8*8  = 64
9*1  = 9        9*2  = 18       9*3  = 27       9*4  = 36       9*5  = 45       9
*6  = 54        9*7  = 63       9*8  = 72       9*9  = 81
```

In [ ]:

```
#打印乘法表
for i in range(1,10):
    for s in range(1,i+1):
        print(f"{i}*{s}  = {i*s}",end='\t')
    print()
```

```
1*1  = 1
2*1  = 2        2*2  = 4
3*1  = 3        3*2  = 6        3*3  = 9
4*1  = 4        4*2  = 8        4*3  = 12       4*4  = 16
5*1  = 5        5*2  = 10       5*3  = 15       5*4  = 20       5*5  = 25
6*1  = 6        6*2  = 12       6*3  = 18       6*4  = 24       6*5  = 30       6
*6  = 36
7*1  = 7        7*2  = 14       7*3  = 21       7*4  = 28       7*5  = 35       7
*6  = 42        7*7  = 49
8*1  = 8        8*2  = 16       8*3  = 24       8*4  = 32       8*5  = 40       8
*6  = 48        8*7  = 56       8*8  = 64
9*1  = 9        9*2  = 18       9*3  = 27       9*4  = 36       9*5  = 45       9
*6  = 54        9*7  = 63       9*8  = 72       9*9  = 81
```

数据类型：列表list

In [ ]:
```python
ages = [11,22,33,44]
print(ages)
#空列表
ages1 = list()
print(ages1)
names = ['lihua',66,False,9.9,ages]#列表可以放不同数据
print(names)
```

```
[11, 22, 33, 44]
[]
['lihua', 66, False, 9.9, [11, 22, 33, 44]]
```

In [ ]:
```python
scores = [100]
scores.append(90)   #添加数据
print(scores)       # 输出 [100, 90]
scores.append(88)   # 先添加元素
print(scores)       # 再打印列表，输出 [100, 90, 88]
scores.insert(0,60) #插入单个元素
print(scores)
ages = [11,22,33,44]
scores.extend(ages) #插入多个元素，列表嵌套
print(scores)
del scores[1]       #根据索引删除元素
print(scores)
scores.pop()
print(scores)
scores.remove(22)   #删除指定元素
print(scores)
scores.sort()       #升序排序
print(scores)
e=scores[2]         #查询
print(e)
scores.extend([2,6,10,15,24,37])
print(len(scores))  #列表长度
scores.sort(reverse=True)       #降序排序
print(scores)
print(scores[2:])
print(scores[2:5])   #切片
print(scores[::2])
```

```
[100, 90]
[100, 90, 88]
[60, 100, 90, 88]
[60, 100, 90, 88, 11, 22, 33, 44]
[60, 90, 88, 11, 22, 33, 44]
[60, 90, 88, 11, 22, 33]
[60, 90, 88, 11, 33]
[11, 33, 60, 88, 90]
60
11
[90, 88, 60, 37, 33, 24, 15, 11, 10, 6, 2]
[60, 37, 33, 24, 15, 11, 10, 6, 2]
[60, 37, 33]
[90, 60, 33, 15, 10, 2]
```

In [ ]:

```
nums = [90, 88, 60, 37, 33, 24, 15, 11, 10, 6, 2]
print(nums[::2])
print(nums[::-1])#全部逆序
print(nums[3:0:-1])#从第四位开始倒序取三个
print(nums[-1:-5:-1])#倒序取五个
```

```
[90, 60, 33, 15, 10, 2]
[2, 6, 10, 11, 15, 24, 33, 37, 60, 88, 90]
[37, 60, 88]
[2, 6, 10, 11]
```

In [ ]:

```
num1 = [[scores],[nums]]
print(num1)
```

```
['[scores]', '[nums]']
```

In [ ]:

```
#作业1
# 定义列表
bags = ['a', 'b', 'c', 'd']

# 1) 计算 bags[int('3')//11]
res1 = bags[int('3') // 11]
print("第1题结果:", res1)

# 2) 计算 bags[-1]
res2 = bags[-1]
print("第2题结果:", res2)

# 3) 计算 bags[-2]
res3 = bags[-2]
print("第3题结果:", res3)
```

```
a
d
c
```

In [6]:

```
#作业2
# 定义初始列表
bacon = [1.15, 'dog', 13, 'cat', True]
print("初始列表:", bacon)

# 1) bacon.index('cat') → 查找'cat'的索引
index_result = bacon.index('cat')
print("\n1) 'cat' 的索引是:", index_result)

# 2) bacon.append(99) → 末尾添加99
```

```
# 重新定义列表，避免上一步修改影响结果
bacon = [1.15, 'dog', 13, 'cat', True]
bacon.append(99)
print("\n2) 执行 append(99) 后的列表:", bacon)

# 3) bacon.remove('cat') → 删除元素'cat'
# 重新定义列表
bacon = [1.15, 'dog', 13, 'cat', True]
bacon.remove('cat')
print("\n3) 执行 remove('cat') 后的列表:", bacon)
```

```
初始列表: [1.15, 'dog', 13, 'cat', True]

1) 'cat' 的索引是: 3

2) 执行 append(99) 后的列表: [1.15, 'dog', 13, 'cat', True, 99]

3) 执行 remove('cat') 后的列表: [1.15, 'dog', 13, True]
```

In [9]:

```
#作业3
# 定义原始列表
li = ['ethan', 'zoran', 'jim']
print("原始列表:", li)

# a. 计算列表长度并输出
print("a. 列表长度:", len(li))

# b. 列表中追加元素"lucy", 并输出添加后的列表
li.append("lucy")
print("b. 追加lucy后的列表:", li)

# c. 在列表的第1个位置插入元素"Tony", 并输出添加后的列表
li.insert(0, "Tony")
print("c. 第1位插入Tony后的列表:", li)

# d. 修改列表第2个位置的元素为"Kelly", 并输出修改后的列表
li[1] = "Kelly"
print("d. 修改第2位为Kelly后的列表:", li)

# e. 删除列表中的元素"ethan", 并输出修改后的列表
# 注意：此时列表中已经没有"ethan"了，为了完成需求，我们重新基于原始列表操作
li = ['ethan', 'zoran', 'jim']
li.remove("ethan")
print("e. 删除ethan后的列表:", li)

# f. 删除列表中的第2个元素，并输出删除元素后的列表
li = ['ethan', 'zoran', 'jim']
del li[1]
print("f. 删除第2个元素后的列表:", li)
```

```
原始列表: ['ethan', 'zoran', 'jim']
a. 列表长度: 3
b. 追加lucy后的列表: ['ethan', 'zoran', 'jim', 'lucy']
c. 第1位插入Tony后的列表: ['Tony', 'ethan', 'zoran', 'jim', 'lucy']
d. 修改第2位为Kelly后的列表: ['Tony', 'Kelly', 'zoran', 'jim', 'lucy']
e. 删除ethan后的列表: ['zoran', 'jim']
f. 删除第2个元素后的列表: ['ethan', 'jim']
```

In [10]:

```
#作业4
nums = [10, 20, 30, 50, 70, 20]
```

```
# 1) 查询20首次出现的索引（遍历方式）
first_index = -1  # 先默认没找到
for i in range(len(nums)):
    if nums[i] == 20:
        first_index = i
        break  # 找到第一个就退出循环

print("1) 20首次出现的索引:", first_index)

# 2) 查询20出现的所有位置
all_index = []
for i in range(len(nums)):
    if nums[i] == 20:
        all_index.append(i)

print("2) 20出现的所有位置:", all_index)
```

```
1) 20首次出现的索引: 1
2) 20出现的所有位置: [1, 5]
```

In [11]:
```
#作业5
names = ["曹操", "刘备", "关羽", "张飞", "小乔", "诸葛亮"]

for i in range(len(names)):
    print(names[i])
```

```
曹操
刘备
关羽
张飞
小乔
诸葛亮
```

In [12]:
```
#作业6
# 原始嵌套列表
names = [["张飞","刘备","关羽"],["曹操","典韦","司马懿"]]

# 空列表，用来存放最终结果
li = []

# 循环遍历每一个小列表
for item in names:
    # 再循环遍历小列表里的每一个名字
    for name in item:
        li.append(name)

# 输出结果
print(li)
```

```
['张飞', '刘备', '关羽', '曹操', '典韦', '司马懿']
```